On the origin of functions by means of natural selection and other processes

Sigfrid Lundberg

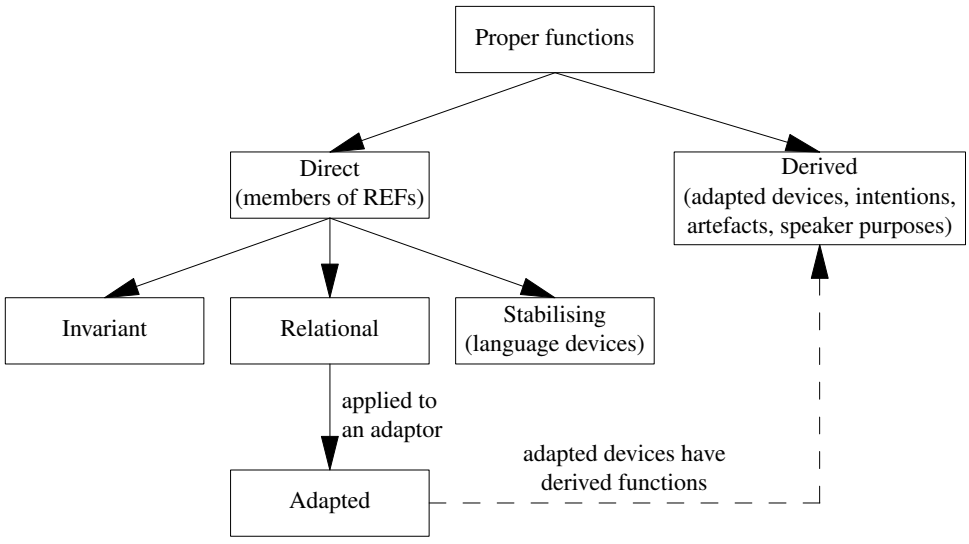
ABSTRACT

I at least try to write my paper before I write my abstract, because I hardly ever know what to summarize before I've written it. This in itself is most likey to be seen as a shortcoming, but be that as it may.

Introduction

If we are to believe Amie L. Thomasson, then many ontological and metaphysical theories have passed their best-before dates. Her arguments are that current ontology is far too complex (Thomasson, 2016) and that metaphysics itself is basically broken (Thomasson, 2026), and that it therefore must be made *functional* again. By functional she does not mean that someone should repair metaphysics but that the old quinean ontology (Quine, 1948) should be replaced with one that don't list existing things but investigate their functions.

> a theory is committed to those and only those entities to which the bound variables of the theory must be capable of referring in order that the affirmations made in the theory be true.



having a cryoprotectant in ones body tissue is similar to have antifreeze in ones coolant

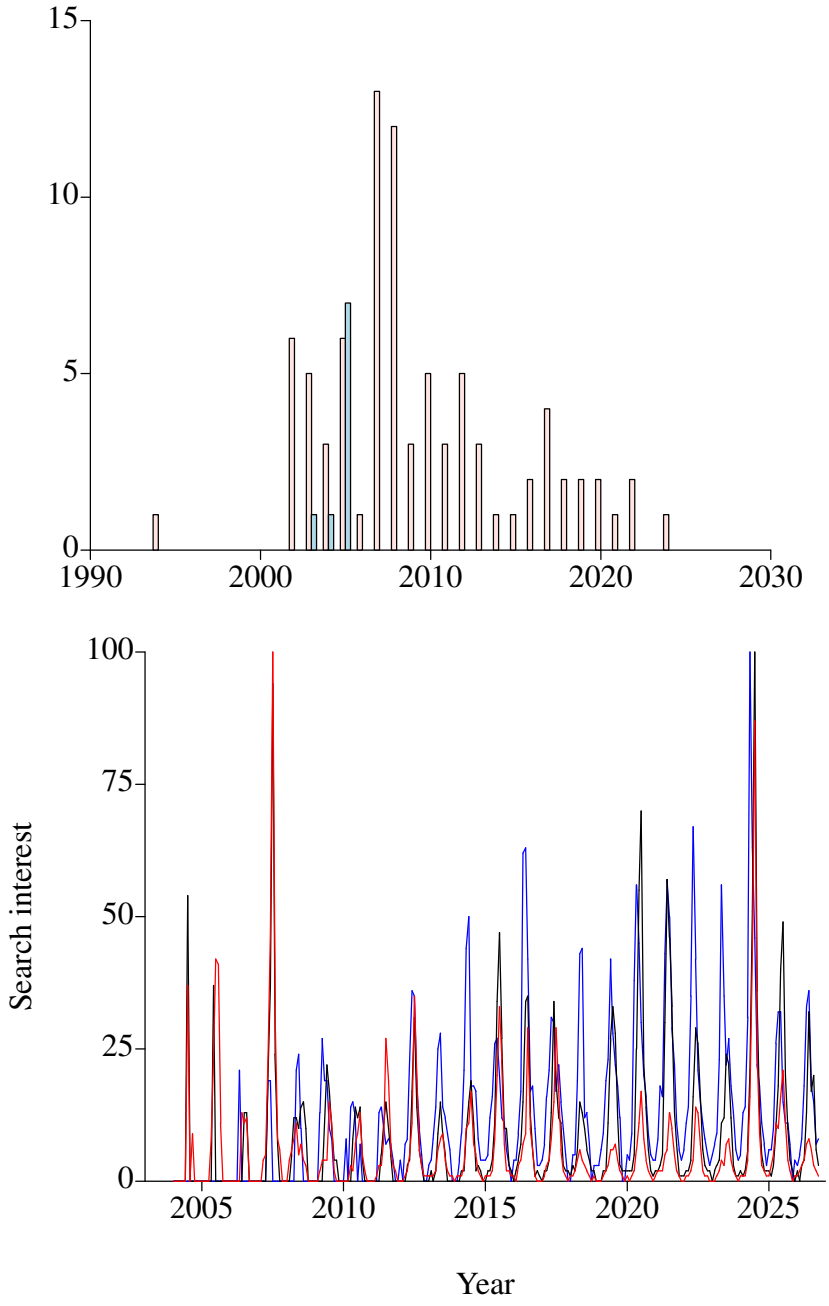
1. The Glucose Accumulation Disparity

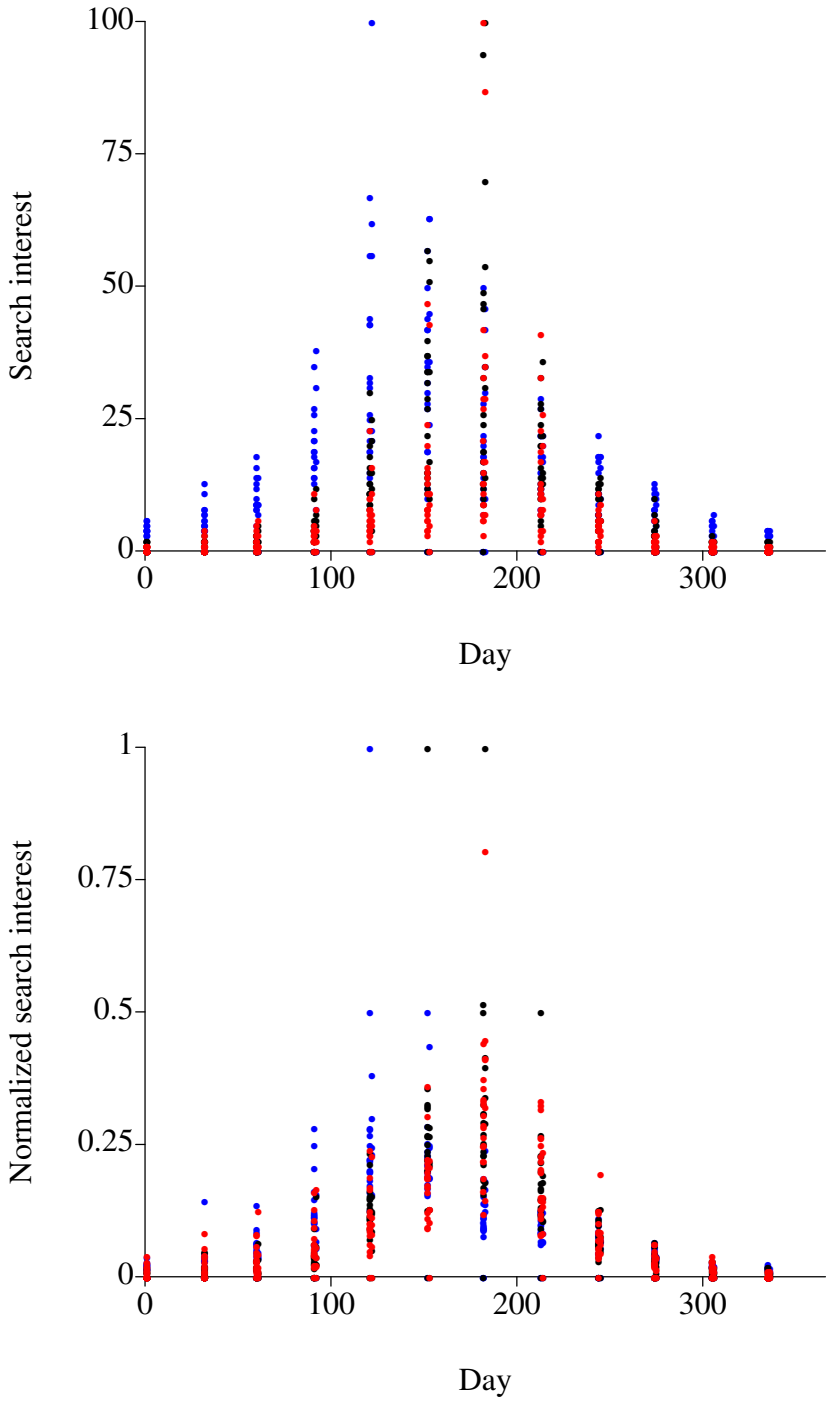
Both species trigger glycogen-to-glucose conversion dynamically upon freezing, but their capacity to ramp up glucose levels differs. • Arion rufus and Arion ater

complexes demonstrate a more pronounced hyperglycemic response to freezing than Arion vulgaris. • Arion vulgaris accumulates glucose upon freezing, but it does so less efficiently and in lower concentrations compared to its cold-adapted northern relatives. Because of this lower peak concentration, it gets less colligative protection from its sugar reserves.

2. Lower Freezing Survival Limits

Because Arion vulgaris originated from more temperate regions of Europe, its internal machinery is less equipped for prolonged deep freezes. • Survival window: While both species can survive short-term freezing (such as 2 days at a mild -1°C), Arion ater handles prolonged or slightly lower temperatures better. In contrast, Arion vulgaris experiences much higher mortality rates if tissue ice content becomes too high or if temperatures drop to even -2°C for an extended period.



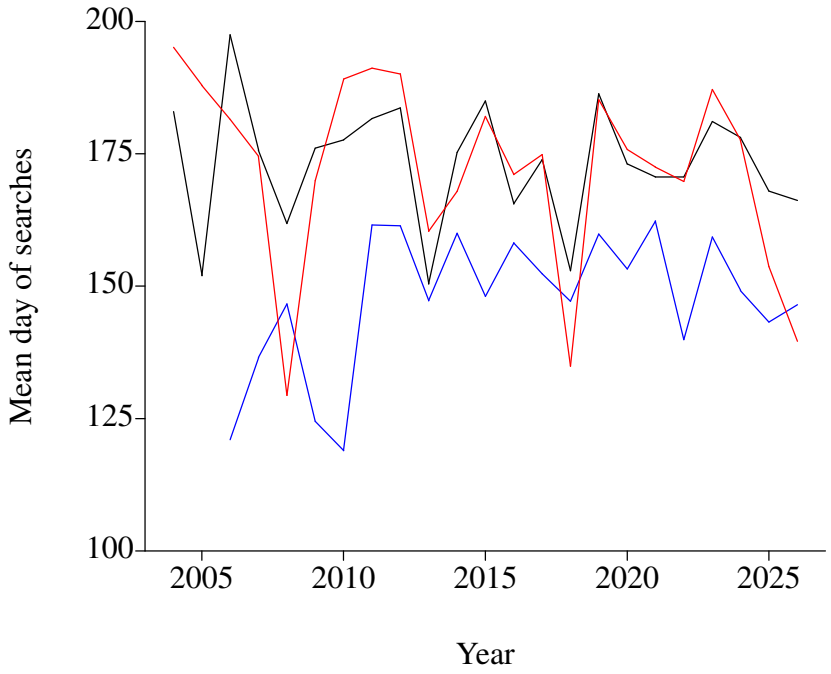


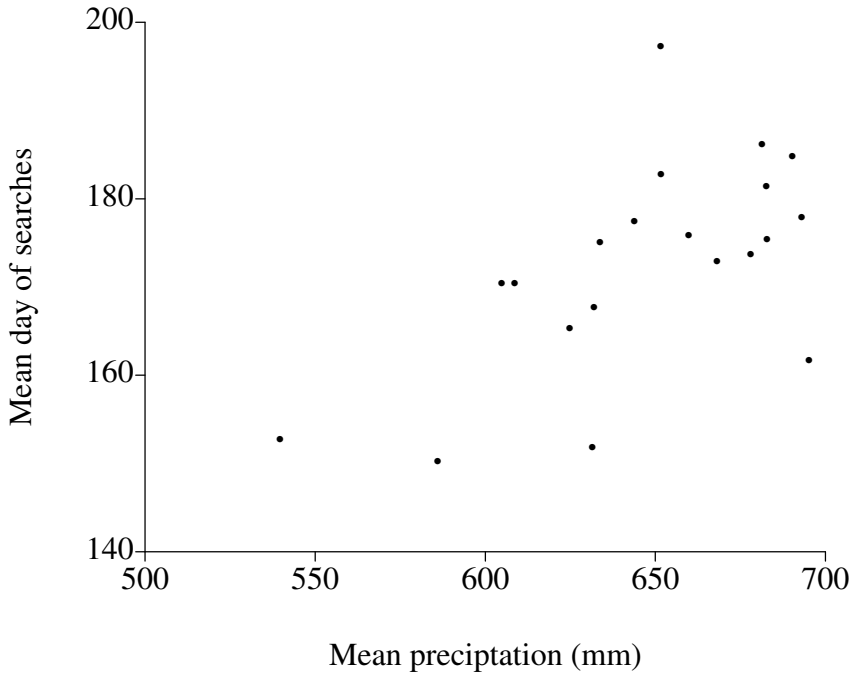
References

Quine, Willard V., "On what there is," *The Review of Metaphysics* **2**, pp. 21-38 (1948). https://www.jstor.org/stable/20123117.

Thomasson, Amie L., "Easy Ontology and the Work of Metaphysics" in *Metaphysics,* Royal Institute of Philosophy, London (2016). https://youtu.be/AbMtCIbBZNU.

Thomasson, Amie L., "Metaphysics Has Stopped Working—Making Metaphysics Functional" in *Making Metaphysics Functional,* Ergo (2026). https://youtu.be/Pl-9tD6qgEo.


200
175
150
125
100
Mean day of searches
2005
2010
2015
2020
2025
Year


200
180
160
140
Mean day of searches
500
550
600
650
700
Mean preciptation (mm)